Quiz02 Practice Questions

Stat 230 - Spring 2026

Practice questions for each skill on the Quiz02 Review sheet. The data sets are ones we've seen in class, but the models and questions are new.

1. Express a MLR model using proper notation

Example question The `bikes` data include the `year` of each day (2011 or 2012). The model below allows the relationship between temperature (`temp_actual`, in degrees F) and daily `rides` to be different in the two years.

```
year_mod <- lm(rides ~ temp_actual * year, data = bikes)
summary(year_mod)
```

```
Call:
lm(formula = rides ~ temp_actual * year, data = bikes)

Residuals:
    Min      1Q  Median      3Q     Max
-4400.9  -565.7     9.0   663.2  2350.4

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)          -1946.022    370.162  -5.257 2.18e-07 ***
temp_actual             71.587      5.833  12.272  < 2e-16 ***
year2012               193.113    539.201   0.358   0.7204
temp_actual:year2012    24.174      8.406   2.876   0.0042 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 951.9 on 496 degrees of freedom
```

```
Multiple R-squared:  0.6368,    Adjusted R-squared:  0.6346
F-statistic: 289.9 on 3 and 496 DF,  p-value: < 2.2e-16
```

(a) Using proper statistical notation, write down the theoretical model that was fit. Write it as a single equation using x_1, x_2, etc., define all variables, and state the assumption on the error term.

(b) Write down the fitted regression equation for 2011 and for 2012 separately.

Example question In general, higher temperatures means more bike rides. But, on very hot days, people may be less likely to bike. This relationship can be captured with a quadratic model. Below is a quadratic model for daily `rides` using temperature.

```
quad_mod <- lm(rides ~ temp_actual + I(temp_actual^2), data = bikes)
tidy(quad_mod)
```

```
# A tibble: 3 x 5
  term             estimate std.error statistic   p.value
  <chr>               <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)      -8708.    2316.         -3.76 0.000190
2 temp_actual        301.      74.2         4.05 0.0000589
3 I(temp_actual^2)    -1.67     0.584      -2.86 0.00444
```

(a) Write down the theoretical model, defining all variables and stating the assumption on the error term.

(b) Write down the fitted model.

(c) Is this still a *linear* regression model, even though the fitted model is a curve? Explain.

2. Interpret the coefficients and R^2 of a model in context

Example question Below is a model for daily `rides` using temperature (`temp_actual`, in degrees F), `humidity` (in %), and whether the day was a weekend (`weekend`).

```
humid_mod <- lm(rides ~ temp_actual + humidity + weekend, data = bikes)
summary(humid_mod)
```

```
Call:
lm(formula = rides ~ temp_actual + humidity + weekend, data = bikes)

Residuals:
```

```
    Min      1Q  Median      3Q     Max
-4462.4  -883.6  -104.1   811.0  2720.7

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -871.252    369.446  -2.358   0.0187 *
temp_actual   97.351      5.465  17.814  < 2e-16 ***
humidity     -25.366      3.628  -6.993 8.77e-12 ***
weekendTRUE -717.710    119.105  -6.026 3.28e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1195 on 496 degrees of freedom
Multiple R-squared:  0.4273,    Adjusted R-squared:  0.4239
F-statistic: 123.4 on 3 and 496 DF,  p-value: < 2.2e-16
```

(a) Interpret the coefficient for `humidity` in context.

(b) Interpret the coefficient for `weekendTRUE` in context.

(c) Interpret the R^2 value in context.

Example question Below is a model for the `Price` of used cars (in dollars) using `Mileage` and the body `Type` of the car (convertible, coupe, hatchback, sedan, or wagon).

```
type_mod <- lm(Price ~ Mileage + Type, data = cars)
summary(type_mod)
```

```
Call:
lm(formula = Price ~ Mileage + Type, data = cars)

Residuals:
   Min     1Q Median     3Q    Max
-12573  -5797  -2292   3689  26652

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.489e+04  1.354e+03  33.158  < 2e-16 ***
Mileage       -1.960e-01  3.489e-02  -5.616  2.7e-08 ***
TypeCoupe     -2.327e+04  1.334e+03 -17.443  < 2e-16 ***
TypeHatchback -2.698e+04  1.551e+03 -17.391  < 2e-16 ***
TypeSedan     -1.996e+04  1.202e+03 -16.596  < 2e-16 ***
```

```
TypeWagon     -1.800e+04  1.528e+03 -11.777  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8096 on 798 degrees of freedom
Multiple R-squared:  0.3334,    Adjusted R-squared:  0.3292
F-statistic: 79.82 on 5 and 798 DF,  p-value: < 2.2e-16
```

(a) Which type of car is the reference level? How can you tell?

(b) Interpret the coefficient for `TypeSedan` in context.

(c) The `Mileage` coefficient is very small because it describes a 1-mile change. Interpret it for an increase of 10,000 miles.

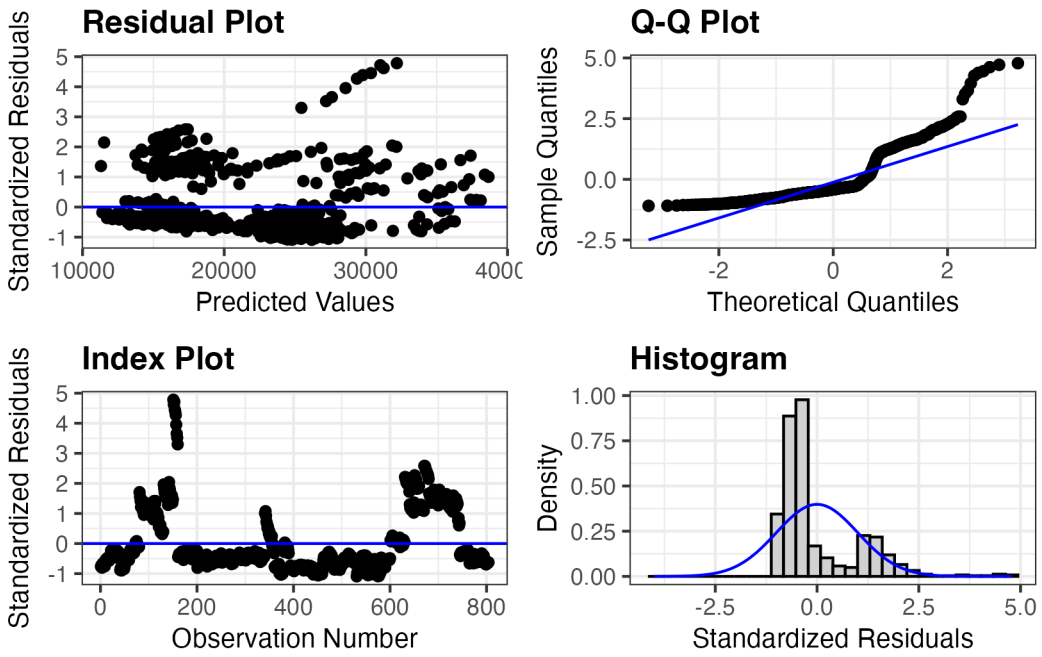
(d) Which of the following is the best explanation for why the adjusted R^2 (0.329) is smaller than the R^2 (0.333)?

A. Adjusted R^2 removes outliers before computing the proportion of variability explained. B. Adjusted R^2 includes a penalty for the number of predictors in the model. C. Adjusted R^2 only uses the quantitative predictors. D. Adjusted R^2 is computed on the log scale.

3. Check necessary conditions for a multiple regression model and discuss the implications of a model deficiency

Example question Here are the residual plots for a model of used car `Price` using `Mileage` and engine size (`Liter`). In this data set, the cars are listed in order of make and model (e.g., all of the Buick Centurys are listed together).

```
liter_mod <- lm(Price ~ Mileage + Liter, data = cars)
resid_panel(liter_mod, type = "standardized")
```



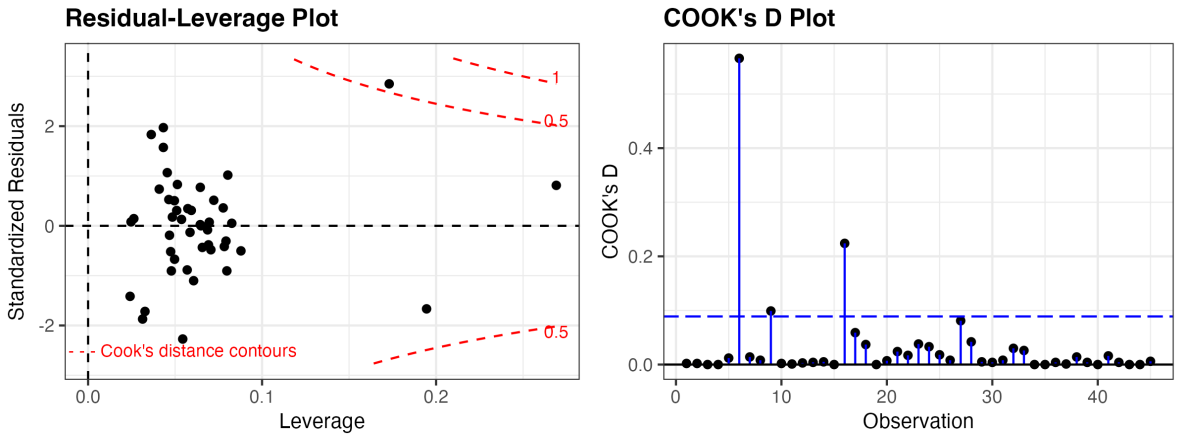
(a) Which condition(s) look violated? Say which plot(s) you used to tell.

(b) Look at the index plot. What do you think is causing the "runs" of residuals that are all above (or all below) zero? What does this suggest about the model?

(c) Given these violations, can you trust the p-values in the `summary()` output for this model? Can you trust the estimated coefficients? Explain.

Example question The `Duncan` data recorded the `prestige`, `income`, and `education` ratings of 45 occupations (all measured as percentages). Here are the leverage and Cook's distance plots, plus the five occupations with the largest Cook's distances, for the model `prestige ~ income + education`.

```
duncan_mod <- lm(prestige ~ income + education, data = duncan)
resid_panel(duncan_mod, plots = c("lev", "cookd"), type = "standardized", nrow = 1)
```




```
augment(duncan_mod) |>
  mutate(occupation = duncan$occupation) |>
  select(occupation, prestige, income, education, .hat, .std.resid, .cooksd) |>
  arrange(desc(.cooksd)) |>
  head(5)
```

```
# A tibble: 5 x 7
  occupation  prestige income education   .hat .std.resid .cooksd
  <chr>          <int>  <int>     <int>  <dbl>      <dbl>   <dbl>
1 minister          87     21        84 0.173       2.85   0.566
2 conductor         38     76        34 0.195      -1.67   0.224
3 reporter          52     67        87 0.0544     -2.27   0.0990
4 RR.engineer       67     81        28 0.269       0.812  0.0810
5 contractor        76     53        45 0.0433      1.97   0.0585
```

(a) Calculate the two leverage cutoffs we discussed in class.

(b) Which occupation has the highest leverage? Is it influential? Explain why or why not.

(c) Which occupation is the most influential? Explain using its leverage and standardized residual.

(d) Below are the coefficients with and without that occupation. Following the flowchart from Day 13, what would you do next?

term	all 45 occupations	without influential point
(Intercept)	-6.065	-6.628
income	0.599	0.732
education	0.546	0.433

4. Perform hypothesis tests about a single regression coefficient

Example question Use the `year_mod` output from Skill 1. Is there evidence that the relationship between temperature and ridership changed from 2011 to 2012?

(a) State the null and alternative hypotheses in words and using proper notation.

(b) Show how the test statistic was calculated, and give its degrees of freedom.

(c) Report the p-value and state your conclusion in context.

Example question Use the `humid_mod` output from Skill 2. Before looking at the data, an analyst hypothesized that more humid days have *fewer* rides, after accounting for temperature and whether it's a weekend.

(a) State H_0 and H_A in symbols, matching the analyst's hypothesis.

(b) Calculate the test statistic.

(c) Is the p-value in the `summary()` output the right one to use? If not, how would you find the correct p-value? State your conclusion.

5. Construct and interpret confidence intervals for regression coefficients and linear combinations of regression coefficients

Example question Use the `humid_mod` output from Skill 2, and the `qt()` output below.

```
qt(.01, df = 496, lower.tail = FALSE)
```

```
[1] 2.333889
```

```
qt(.025, df = 496, lower.tail = FALSE)
```

```
[1] 1.964758
```

```
qt(.05, df = 496, lower.tail = FALSE)
```

```
[1] 1.647932
```

```
qt(.1, df = 496, lower.tail = FALSE)
```

```
[1] 1.283261
```

(a) Construct a 90% confidence interval for the coefficient of `humidity`.

(b) Interpret your interval in context.

Example question Use the `year_mod` model from Skill 1. Below is the covariance matrix of the estimated coefficients.

```
vcov(year_mod)
```

```
                     (Intercept) temp_actual    year2012 temp_actual:year2012
(Intercept)           137019.556 -2129.66456 -137019.556           2129.66456
temp_actual             -2129.665    34.02686    2129.665            -34.02686
year2012              -137019.556  2129.66456  290737.973          -4475.41705
temp_actual:year2012     2129.665   -34.02686   -4475.417             70.66346
```

```
qt(.025, df = 496, lower.tail = FALSE)
```

```
[1] 1.964758
```

(a) Write the slope for temperature *in 2012* as a linear combination of the model coefficients, and calculate its estimate.

(b) Calculate the standard error of the 2012 slope. Why can't you just add the two standard errors from the `summary()` output?

(c) Construct a 95% confidence interval for the 2012 slope, and interpret it in context.

(d) A city planner claims that, in 2012, each additional degree F was associated with 100 more rides per day, on average. Conduct a hypothesis test of this claim.

6. Use a regression model to make appropriate predictions and communicate the uncertainty of these predictions through an interval

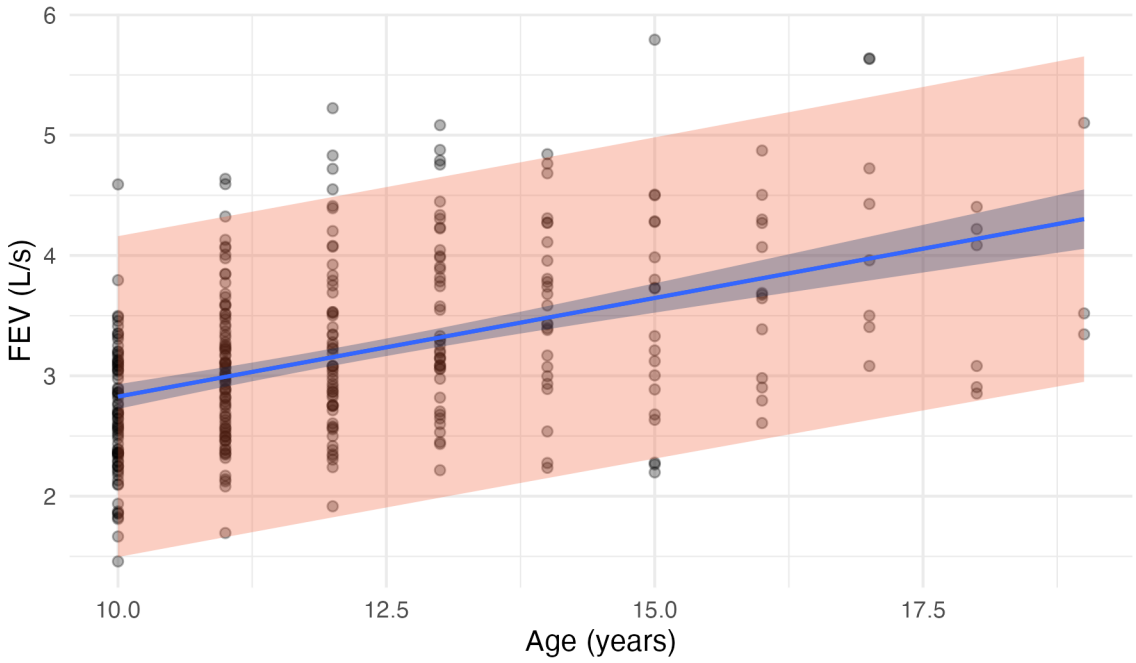
Example question Use the `humid_mod` model from Skill 2. For a weekend day with a temperature of 80°F and 50% humidity, the predicted number of rides is 4931. R gives these two 95% intervals: (2563, 7299) and (4628, 5233).

(a) Which interval is the confidence interval and which is the prediction interval? How can you tell?

(b) Capital Bikeshare is scheduling staff for next Saturday, which is forecast to be 80°F with 50% humidity. Which interval should they use?

(c) Interpret the interval you did *not* choose in (b).

Example question The plot below shows lung function (`FEV`, in L/s) vs. `Age` for the 345 adolescents in the FEV data, along with the fitted SLR line and two shaded bands.



(a) Which band (wide orange or narrow dark) shows the confidence intervals, and which shows the prediction intervals?

(b) Why does it make sense that one band is so much wider than the other?

(c) Where is the narrow dark band narrowest? Why?

7. Conduct inference to compare nested regression models

Example question In the first week of class, we used the NHANES data to model a depression risk score (`dep_score`, the log of the PHQ-9 score + 1) using daily sitting time (`sedentary_hours`). The data also record each participant's `sex`. Researchers want to know whether sex is related to the depression score *at all* (as either a shift in the line, a change in the slope, or both), after accounting for sitting time. They compare these two models:

- Model 1: `dep_score ~ sedentary_hours`
- Model 2: `dep_score ~ sedentary_hours * sex`

Model	Residual df	SSE
Model 1: dep_score ~ sedentary_hours	657	512.88

Model	Residual df	SSE
Model 2: dep_score ~ sedentary_hours * sex	655	503.90

(a) Are these models nested? Which is the full model and which is the reduced model?

(b) Write out the full model, then state the null and alternative hypotheses using proper β notation.

(c) Calculate the F-statistic and give its degrees of freedom.

(d) Describe how the p-value is found (what distribution, and which tail?). The output below may help.

```
pf(5.83, df1 = 2, df2 = 655, lower.tail = FALSE)
```

```
[1] 0.003092683
```

(e) What do you conclude?

Example question Consider these four models for the `bikes` data:

- Model A: `rides ~ temp_actual`
- Model B: `rides ~ temp_actual + humidity`
- Model C: `rides ~ temp_actual * year + humidity`
- Model D: `rides ~ humidity + weekend`

For each comparison, say whether the most appropriate test is a t-test, an extra-sums-of-squares F-test, or neither, and briefly explain.

(a) Model A vs. Model B

(b) Model A vs. Model C

(c) Model B vs. Model D

(d) The `summary()` output for Model C ends with an F-statistic. Which two models does that F-test compare?

8. Interpret the coefficients an MLR model when transformations have been used on the response and/or the explanatory variables

Example question Below is a model for the log of used car prices using engine size (`Liter`) and whether the car has cruise control (`Cruise`: 1 = yes, 0 = no).

```
cruise_mod <- lm(log(Price) ~ Liter + Cruise, data = cars)
tidy(cruise_mod)
```

```
# A tibble: 3 x 5
  term        estimate std.error statistic  p.value
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)    9.12     0.0329     277.  0
2 Liter          0.175    0.0106      16.4 2.02e-52
3 Cruise         0.300    0.0273      11.0 2.63e-26
```

(a) Interpret the `Liter` coefficient in terms of price on the original scale (dollars).

(b) Interpret the `Cruise` coefficient in terms of price.

Example question Here are two models for used car `Price` using a log transformation of `Mileage`.

```
logx_mod <- lm(Price ~ log(Mileage), data = cars)
coef(logx_mod)
```

```
 (Intercept) log(Mileage)
   43013.577    -2221.761
```

```
loglog_mod <- lm(log(Price) ~ log(Mileage), data = cars)
coef(loglog_mod)
```

```
 (Intercept) log(Mileage)
10.72537349  -0.08676928
```

(a) For `logx_mod`, interpret the slope in terms of a *doubling* of mileage.

(b) For `loglog_mod`, which of the following is the best interpretation of the slope?

A. A 1-mile increase in mileage is associated with a 0.087 dollar decrease in price, on average. B. A 1% increase in mileage is associated with an estimated 0.087% decrease in price. C. A 1% increase in mileage is associated with an estimated 8.7% decrease in price. D. A 1-mile increase in mileage is associated with an $e^{-0.087}$ dollar decrease in price, on average.

(c) Suppose we had used $\sqrt{\text{Price}}$ as the response instead. Could we interpret the slope on the original scale (dollars) the same way? Why or why not?

9. Diagnose multicollinearity and explain its consequences

Example question Below is a model for the log of used car prices using `Mileage`, engine size (`Liter`), and the number of cylinders (`Cyl`).

```
cyl_mod <- lm(log(Price) ~ Mileage + Liter + Cyl, data = cars)
tidy(cyl_mod)
```

```
# A tibble: 4 x 5
  term           estimate  std.error statistic    p.value
  <chr>             <dbl>      <dbl>     <dbl>      <dbl>
1 (Intercept)  9.26       0.0651        142.   0
2 Mileage     -0.00000674 0.00000140     -4.80 0.00000189
3 Liter        0.148      0.0363          4.09 0.0000468
4 Cyl          0.0579     0.0289          2.00 0.0454
```

```
cars |>
  select(Mileage, Liter, Cyl) |>
  cor()
```

```
             Mileage       Liter         Cyl
Mileage  1.00000000 -0.01864062 -0.02946099
Liter   -0.01864062  1.00000000  0.95789658
Cyl     -0.02946099  0.95789658  1.00000000
```

```
car::vif(cyl_mod)
```

```
  Mileage     Liter       Cyl
 1.001985 12.144428 12.150755
```

```
summary(lm(Liter ~ Mileage + Cyl, data = cars))$r.squared
```

```
[1] 0.9176577
```

For comparison, here is the same model without `Cyl`:

```
tidy(lm(log(Price) ~ Mileage + Liter, data = cars))
```

```
# A tibble: 3 x 5
  term           estimate  std.error statistic  p.value
  <chr>             <dbl>      <dbl>     <dbl>    <dbl>
1 (Intercept)  9.35       0.0441        212.   0
2 Mileage     -0.00000686 0.00000141     -4.88 1.30e- 6
3 Liter        0.218      0.0104         20.9  7.77e-78
```

(a) Which pair of predictors is the most problematic in terms of multicollinearity? Why?

(b) Show how the VIF for `Liter` was calculated.

(c) Using the guidelines from class, how serious is the multicollinearity in this model?

(d) Compare the estimate and standard error for `Liter` in the two models. What happened, and why?

(e) Suppose we only want to use `cyl_mod` to *predict* the price of used cars like the ones in this data set. Is the multicollinearity a big problem for that goal? Explain.